JAVASCRIPT ASSIGNMENT : 14

Module 14: ES6 Plus Features – Multiple Assignment Questions

Section 1: let and const

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1. Demonstrate block scope using let

Create example where variable declared inside block using let is not accessible outside.
Hint: Compare with var.

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2. Show const behavior with object

Create object using const and modify its property.
Hint: Reassignment not allowed but property update allowed.

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3. Create loop example comparing var and let

Create loop and print values using setTimeout inside loop.
Hint: Observe closure difference.

Section 2: Arrow Functions

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4. Convert normal function into arrow function

Take simple addition function and convert into arrow syntax.
Hint: Use concise syntax.

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5. Create arrow function returning object

Create arrow function that returns object with name and age.
Hint: Wrap object in parentheses.

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6. Compare this in arrow and normal function

Create object with method using arrow and normal function and observe difference.
Hint: Arrow does not bind its own this.

Section 3: Destructuring

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7. Destructure object properties

Create object and extract properties into variables using destructuring.
Hint: Variable names must match keys.

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8. Rename variables during destructuring

Destructure object and rename properties.
Hint: Use colon syntax.

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9. Destructure nested object

Create nested object and extract nested values directly.
Hint: Destructure inside destructuring.

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10. Destructure array elements

Create array and extract first two elements into variables.
Hint: Use index based destructuring.

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11. Swap two variables using destructuring

Swap values of two variables without temporary variable.
Hint: Use array destructuring.

Section 4: Spread Operator

12. Merge two arrays using spread

Create two arrays and merge them into one.
Hint: Use spread syntax.

13. Clone object using spread

Clone object and modify cloned object without affecting original.
Hint: Spread creates shallow copy.

14. Merge two objects

Combine two objects into one object using spread.
Hint: Later properties override earlier ones.

Section 5: Rest Operator

15. Create function using rest parameters

Create function that accepts multiple numbers and returns sum.
Hint: Rest collects parameters into array.

16. Separate first element and rest elements from array

Create array and destructure first element separately and rest in another array.
Hint: Use rest in destructuring.

Section 6: Optional Chaining

17. Access deeply nested property safely

Create nested object and access property using optional chaining.
Hint: Prevent undefined errors.

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18. Demonstrate difference with and without optional chaining

Create example where accessing undefined property causes error and fix using optional chaining.
Hint: Compare outputs.

Section 7: Nullish Coalescing

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19. Provide default value using nullish coalescing

Create variable and assign default value if it is null or undefined.
Hint: Use double question mark operator.

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20. Compare logical OR and nullish coalescing

Create example where value zero or empty string is handled differently.
Hint: Understand falsy vs nullish.

Section 8: Real World Combined Questions

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21. Create user profile handler

Create function that accepts user object and safely prints user details using destructuring and optional chaining.
Hint: Provide default values.

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22. Create shopping cart merge system

Merge default cart items and user selected items using spread operator.
Hint: Combine arrays.

23. Create flexible calculator using rest and arrow functions

Create arrow function that accepts unlimited numbers and returns sum.
Hint: Use rest parameter.

24. Create safe API response handler

Simulate API response object and access nested data safely using optional chaining.
Hint: Avoid runtime error.

25. Create configuration system with default values

Create configuration object and assign default values using nullish coalescing.
Hint: Handle undefined properties.

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